

Do Deep EEG Networks Really Not Need Preprocessing? A Matched Comparison of No-Preprocessing and Preprocessed EEG for Four-Class Motor-Imagery Decoding

Abstract—A frequent argument for deep learning in motor-imagery (MI) brain-computer interfaces (BCIs) holds that convolutional neural networks (CNNs) can discover discriminative features straight from no-preprocessing electroencephalography (EEG), rendering classical signal preprocessing unnecessary. I put that argument to the test under a strictly matched protocol on the four-class BCI Competition IV dataset 2a. Two widely used architectures, Deep4Net and ShallowFBCSPNet, each receive two versions of the identical trials: a lightly processed *no-preprocessing* pipeline and a *preprocessed* pipeline that adds 8–30 Hz filtering together with artifact cleaning. Inside each architecture the no-preprocessing and preprocessed variants share one common recipe (cross-subject pretraining, subject-specific fine-tuning, crop-level decision fusion and every hyper-parameter), so any difference between them traces back to the input representation alone. Pooled over nine subjects, preprocessing nudges mean accuracy up by only 1.5–2.2 percentage points for either model but, more tellingly, contracts the between-subject spread: the accuracy standard deviation of ShallowFBCSPNet falls from 20.40 to 15.11 and its poorest subject climbs from near chance (26.45%) up to 49.47%. These improvements cluster on the toughest subjects, those decoding worst from no-preprocessing input. Overall, deep EEG models clearly do recover usable structure from unfiltered signals, yet domain-informed preprocessing keeps its worth chiefly as a *variance-reduction mechanism* that props up the weakest subjects rather than as a sizeable average boost.

Index Terms—Brain-computer interface, motor imagery, EEG, convolutional neural network, Euclidean Alignment, transfer learning, preprocessing.

I. INTRODUCTION

A brain-computer interface (BCI) links neural activity, captured non-invasively through electroencephalography (EEG), to an external device without depending on muscular output [1], [2]. Motor imagery (MI), mentally rehearsing a movement while remaining physically still, is among the most extensively studied BCI paradigms, since it elicits characteristic amplitude changes in the μ (8–13 Hz) and β (13–30 Hz) sensorimotor rhythms that a decoder can map to intended commands [3]. Dependable MI decoding underpins uses spanning neurorehabilitation, assistive robotics and communication support for people with limited motor function [2].

For much of the field’s history, MI pipelines paired engineered spatial filters (most notably Common Spatial Patterns and their filter-bank variant) with linear classifiers [6]. These pipelines perform well yet remain fragile to noise and to the shifting statistics of EEG across days and individuals [4], [5]. Deep learning has since narrowed this gap: convolutional models learn their temporal and spatial filters directly from data and now rival or exceed the engineered baselines [7]–[9], while convolution–transformer hybrids additionally capture longer-range temporal structure [10]–[12].

A recurring selling point of the deep-learning route is that it does away with expert preprocessing: supplied with enough no-preprocessing data, the network is said to uncover the relevant frequency bands and channels unaided. The claim is appealing but seldom examined in a controlled way, since published comparisons typically alter the architecture, the training recipe and the input format all at once [13]. As a result, it stays unclear whether a reported improvement stems from the model itself or from the signal processing wrapped around it.

I isolate the input representation in this study. I hold the architecture, the alignment step, the training schedule, the augmentation and every hyper-parameter constant, and toggle only whether the model sees a lightly processed *no-preprocessing* signal or a *preprocessed* signal confined to the 8–30 Hz motor range with artifacts stripped out. I examine two contrasting networks: Deep4Net, a deep CNN, and ShallowFBCSPNet, whose front end imitates band-power computation [7]. My main contributions are as follows:

- 1) A matched-protocol comparison of no-preprocessing versus preprocessed EEG on BCI Competition IV-2a that, within each architecture, controls for pretraining, fine-tuning and augmentation, so that only the input representation differs.
- 2) Evidence that preprocessing yields modest average gains but a large reduction in cross-subject variance, with the benefit concentrated on the hardest subjects.
- 3) A standardized transfer-learning pipeline combining cross-subject pretraining, low-rate fine-tuning, Mixup, cropped-decision pooling and, for the Deep4Net pipeline,

Euclidean Alignment.

II. DATASET AND PREPROCESSING

A. Dataset

My experiments use BCI Competition IV dataset 2a [14], [15]. It comprises nine subjects (A01–A09), each performing four imagined actions (left hand, right hand, both feet and tongue) recorded over 22 EEG channels at 250 Hz. Every subject provides two sessions captured on different days: a training session (T) and an evaluation session (E), each containing 288 trials (72 per class). Per the competition protocol, I train on session T and reserve session E, together with its officially released labels, purely for testing. Because the two sessions originate from separate days, the resulting scores reflect a realistic cross-session setting rather than an easier within-session estimate. Trials that the recording software flagged as contaminated are excluded.

B. Two Matched Pipelines

Only the input representation sets the two conditions apart. Under the *no-preprocessing* condition, each GDF recording is sliced into 0–4 s post-cue epochs and retained as a $22 \times T$ voltage array with no spectral processing, leaving the model with what is effectively the sensor trace. Under the *preprocessed* condition, those same epochs additionally pass through an 8–30 Hz band-pass that preserves the μ/β motor rhythms, a 50 Hz notch that suppresses mains interference, and FastICA-based ocular cleaning in which components are flagged automatically by their correlation with the three EOG reference channels and discarded. The EOG channels act purely as artifact references and never form part of the 22-channel model input. Critically, the preprocessed output keeps the same $22 \times T$ shape; only its frequency content is altered, which is precisely what makes the two conditions directly comparable. All signal handling uses MNE-Python [16].

III. EUCLIDEAN ALIGNMENT AND CROPPED TRAINING

Fig. 1 lays out the two architecture pipelines side by side. EEG covariance structure shifts considerably from one subject and session to another, which hurts a model trained on pooled recordings. To curb this drift I adopt Euclidean Alignment (EA) [17]. Given a set of trials $\{X_i\}_{i=1}^N$ from a single subject, where each $X_i \in \mathbb{R}^{22 \times T}$ is a 22-channel epoch of length T , EA first forms the mean spatial covariance

$$\bar{R} = \frac{1}{N} \sum_{i=1}^N X_i X_i^\top, \quad (1)$$

and whitens every trial as

$$\tilde{X}_i = \bar{R}^{-1/2} X_i, \quad (2)$$

where $\bar{R}^{-1/2}$ comes from the eigendecomposition of $\bar{R}$, with its eigenvalues clipped to a small positive floor to keep the operation numerically stable. Once aligned, every subject’s trials share an identity mean covariance, which shrinks the between-subject distribution gap and, being fully unsupervised, needs no target-side labels [17], [18]. Here EA is applied inside

the Deep4Net pipeline for both the no-preprocessing and preprocessed inputs, but the ShallowFBCSPNet pipeline is trained without it (Fig. 1(a) vs. (b)). The no-preprocessing-versus-preprocessed contrast thus stays matched within each architecture, whereas the two architectures differ in whether alignment is present; the consequence of this choice is revisited in Section VI.

Every (aligned) trial is then unpacked into overlapping 500-sample windows spaced 50 samples apart, mirroring the cropped-training idea of [7]. This windowing doubles as temporal augmentation and produces many predictions per trial. Per-channel means and standard deviations are computed from the training windows alone and reused across all splits, so no evaluation statistics leak backward. To stop windows from crossing the split boundary, trials are partitioned into training and validation sets *before* windowing.

IV. CNN ARCHITECTURES AND TRAINING

Both models are drawn from the braindecode library [7] and set up for 22 input channels, four output classes and a 500-sample input window with dropout 0.5.

A. Deep4Net

Deep4Net is a four-block deep CNN in which stacked temporal and spatial convolutions, batch normalization and max pooling build up a hierarchy of spectral-spatial features straight from the time series.

B. ShallowFBCSPNet

ShallowFBCSPNet is a shallow model whose squaring-and-log-pooling front end mimics the band-power computation of Filter Bank Common Spatial Patterns, giving a learnable analogue of the classical baseline; its inductive bias therefore presumes reasonably clean band-power inputs.

C. Training Protocol

Training proceeds in two phases that stay identical across conditions for a given model. In the first phase, one shared backbone is pretrained on the pooled T-session windows of all nine subjects (aligned in the Deep4Net case) with a 90/10 trial-level split, Adam as the optimizer, a weight decay of 0.01, a cosine-annealing rate schedule, label-smoothed (0.1) cross-entropy, and Mixup at $\alpha = 0.2$ [19], [24]. This phase lasts up to 100 epochs at a base rate of 10^{-3} , a batch size of 128, and patience-15 early stopping keyed to validation loss. In the second phase, the pretrained weights are tuned to each target subject on that person’s T session (80/20 split) for up to 50 epochs at the lower rate of 10^{-4} and a batch size of 32, once more with Mixup, so the shared representation is gently refined instead of overwritten. A single fixed seed (42) drives all splits and stochastic steps for reproducibility. At inference, the model scores every window of an E-session trial and these window outputs are merged into one trial-level label, either by majority vote or by averaging the softmax probabilities. All models run in PyTorch and are scored with classification accuracy and Cohen’s κ , the four-class chance level being 25%.

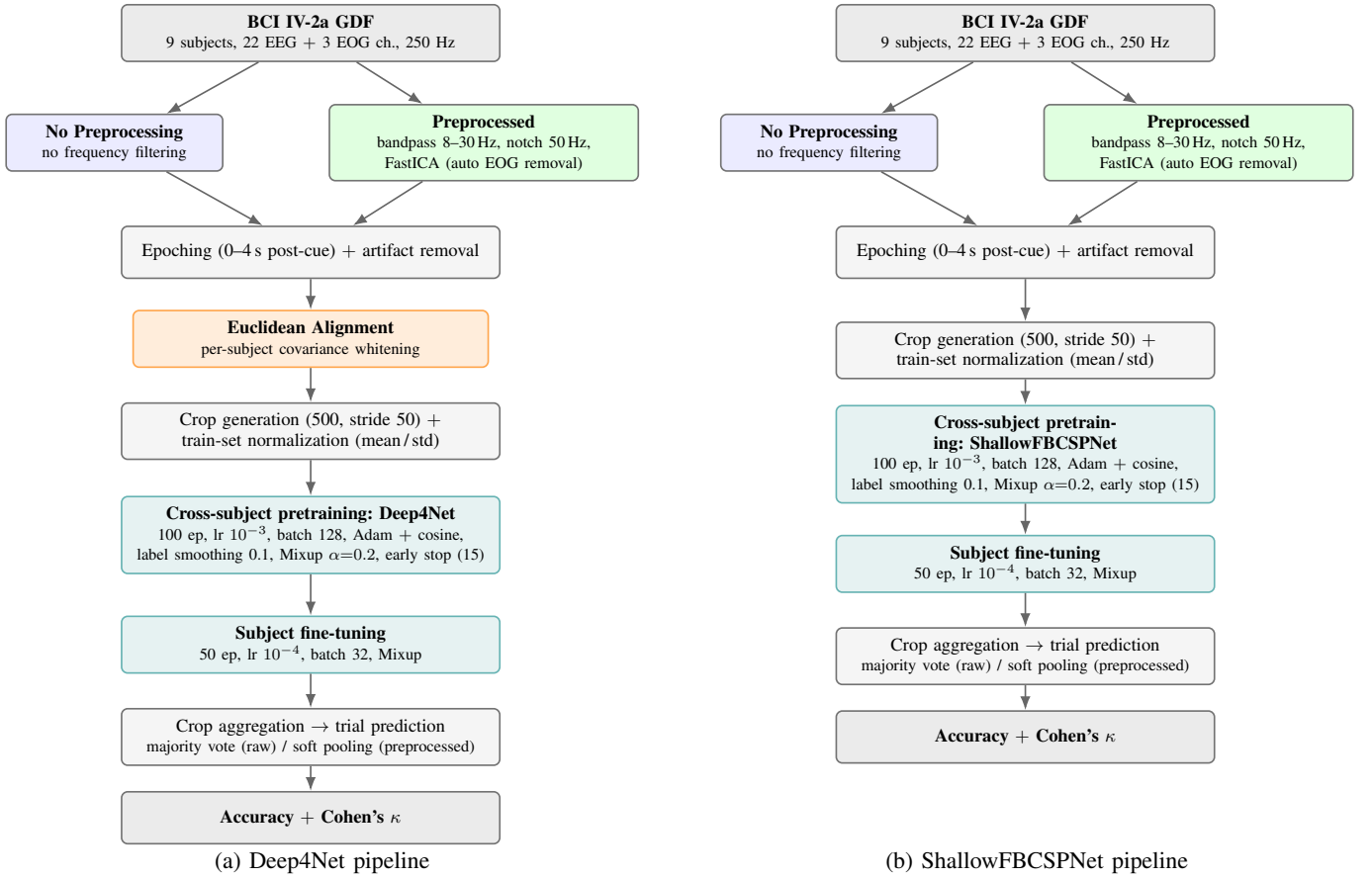


Fig. 1: End-to-end pipelines for the two architectures. Within each panel, the only controlled variable is the input representation: the *no-preprocessing* branch omits frequency filtering, whereas the *preprocessed* branch adds 8–30 Hz band-pass, 50 Hz notch and FastICA artifact removal; all downstream stages are otherwise identical. The two panels differ only in the network and in whether Euclidean Alignment is applied: (a) the Deep4Net pipeline includes it, while (b) the ShallowFBCSPNet pipeline does not.

V. RESULTS

For every (model, pipeline) pairing I execute the complete pretrain-then-fine-tune routine and record per-subject accuracy and κ on the held-out E session. Table I gives the nine-subject summary, and Table II breaks the same accuracies down by subject.

A. Aggregate Comparison

For both architectures, preprocessing raises the mean: Deep4Net climbs from 69.16% to 70.66% and ShallowFBCSPNet from 66.89% to 69.08%, with Cohen’s κ shifting the same way (0.589 $\rightarrow$ 0.609 and 0.561 $\rightarrow$ 0.588). The average benefit is thus genuine but small, on the order of 1.5–2.2 points, in keeping with the notion that a CNN already captures much of the useful structure from the no-preprocessing trace.

B. Statistical Analysis

I assessed the no-preprocessing-versus-preprocessed difference with paired, per-subject comparisons ($n = 9$, $\alpha = 0.05$), combining a paired t -test with the distribution-free Wilcoxon signed-rank test. For Deep4Net the +1.56-point gain fails to reach significance (paired t : $p = 0.30$; Wilcoxon: $p = 0.47$;

TABLE I: Nine-Subject Summary on BCI-IV-2a (E Session). Accuracy in %.

Model	Pipeline	Mean	Std	Min	Max	$\bar{\kappa}$
Deep4Net	No Preprocessing	69.16	14.96	49.64	87.19	0.589
Deep4Net	Preprocessed	70.66	11.79	54.71	85.77	0.609
ShallowFBCSPNet	No Preprocessing	66.89	20.40	26.45	88.09	0.561
ShallowFBCSPNet	Preprocessed	69.08	15.11	49.47	88.28	0.588

95% CI on the mean difference $[-1.7, +4.8]$ points). For ShallowFBCSPNet the +2.22-point gain is similarly non-significant (paired t : $p = 0.53$; Wilcoxon: $p = 0.94$; 95% CI $[-5.6, +10.1]$ points). With just nine subjects the design lacks the power to confirm a small mean shift, and the broad intervals echo the wide spread of the no-preprocessing condition. Far from weakening my argument, these outcomes reinforce it: the worth of preprocessing lies not in a reliable mean gain but in the tightened between-subject spread and the rescue of the weakest subjects, examined next.

TABLE II: Per-Subject Test Accuracy Across All Four Conditions.

Subject	Deep4Net (Preprocessed)	Deep4Net (No Preprocessing)	ShallowFBCSPNet (Preprocessed)	ShallowFBCSPNet (No Preprocessing)
A01	0.8648	0.8399	0.8648	0.8541
A02	0.5724	0.5406	0.4947	0.4311
A03	0.8315	0.8425	0.8828	0.8462
A04	0.6711	0.6184	0.5702	0.5658
A05	0.5181	0.3225	0.5290	0.2645
A06	0.6465	0.5674	0.5488	0.6047
A07	0.8303	0.8628	0.8773	0.8809
A08	0.8266	0.7565	0.7565	0.8118
A09	0.8106	0.7765	0.6932	0.7614

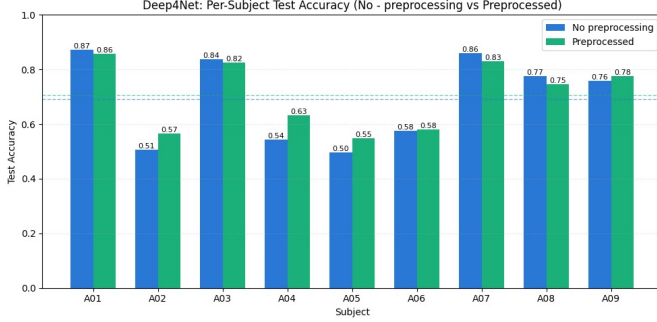


Fig. 2: Deep4Net per-subject test accuracy on the held-out E session, under no-preprocessing versus preprocessed inputs. Dashed lines mark the nine-subject means.

C. The Main Benefit Is Stability

The sharpest effect shows up not in the centre but in the spread. The accuracy standard deviation drops from 14.96 to 11.79 for Deep4Net and from 20.40 to 15.11 for ShallowFBCSPNet, and the minimum rises markedly, most dramatically for ShallowFBCSPNet, whose weakest subject moves from 26.45% (near chance) up to 49.47%. Put differently, preprocessing chiefly makes the decoder *dependable across subjects* rather than lifting the ceiling on the already-easy ones. The κ standard deviations contract under preprocessing for both models, echoing the same pattern.

D. Per-Subject Behavior

Figs. 2 and 3, together with Table II, resolve the results per subject and expose two regimes. “BCI-efficient” subjects such as A01, A03 and A07 already clear 80% on no-preprocessing input, and preprocessing barely moves them. The harder subjects, whose sensorimotor rhythms are fainter or noisier, gain clearly: for ShallowFBCSPNet, A05 leaps from near-chance no-preprocessing accuracy to about 53% once the signal is preprocessed, and several mid-range subjects (A02, A04, A06) shift upward likewise. A handful of easy subjects slip by a point or two under preprocessing, presumably because some useful broadband content is filtered out, yet this modest cost is repaid by the gains on the tougher subjects.

E. Chance-Corrected Agreement

No-preprocessing accuracy can flatter a decoder on the easier subjects, so I repeat the per-subject view using Cohen’s κ , which discounts agreement expected by chance (Fig. 4). The

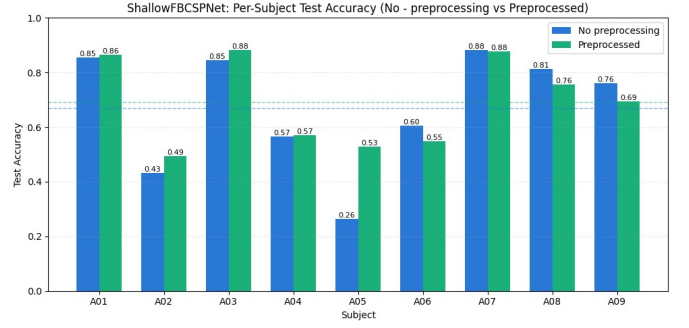


Fig. 3: ShallowFBCSPNet per-subject test accuracy on the held-out E session, under no-preprocessing versus preprocessed inputs. Dashed lines mark the nine-subject means. The gain is largest on the weakest subjects (e.g., A05).

κ profiles track the accuracy profiles closely: the efficient subjects (A01, A03, A07) stay high under either input, while the weak subjects show the clearest lift from preprocessing: for ShallowFBCSPNet, A05 rises from a near-zero κ on no-preprocessing input to roughly 0.37 once filtered. This agreement between the two metrics reinforces the reading that preprocessing chiefly rescues the hardest subjects rather than boosting the already-strong ones.

F. Model Comparison

Read across the table, the Deep4Net pipeline records a higher mean and a smaller spread than the ShallowFBCSPNet pipeline. This cross-architecture reading, however, warrants caution: Deep4Net additionally enjoys Euclidean Alignment, which ShallowFBCSPNet lacks, so the gap between them blends the network with the presence of alignment rather than isolating the architecture. What *is* fully controlled, and hence safe to interpret, is the no-preprocessing-versus-preprocessed contrast *within* each architecture, where preprocessing consistently narrows the spread and elevates the weakest subjects. Even then, the far larger no-preprocessing spread of ShallowFBCSPNet (its worst no-preprocessing subject hovering near chance) signals that its band-power front end is more exposed to noise on unfiltered input, unsurprising, since a shallow band-power model banks on the clean band-power structure that filtering delivers.

G. Combined View

Fig. 5 places all four (model, input) conditions on a single per-subject axis. The joint picture is easy to read: the bars move together on the efficient subjects and separate on the difficult ones, exactly where the choice of input is decisive. No configuration collapses on any individual: each stays above the four-class chance baseline throughout, which confirms that the effects reported above are shifts in an already-working decoder rather than isolated failures.

VI. DISCUSSION

My findings temper the widespread claim that deep networks make preprocessing obsolete. Both models decode the

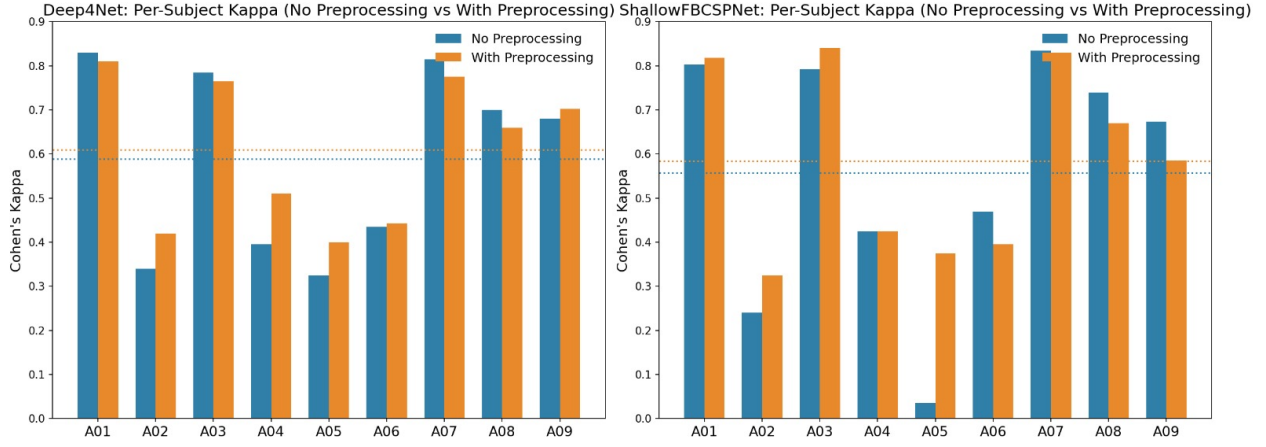


Fig. 4: Per-subject Cohen’s κ on the held-out E session for (left) Deep4Net and (right) ShallowFBCSPNet, under no-preprocessing versus preprocessed inputs. Dashed lines mark the nine-subject means. The chance-corrected view mirrors the accuracy trend, with the sharpest lift on the weakest subjects (e.g., ShallowFBCSPNet on A05).

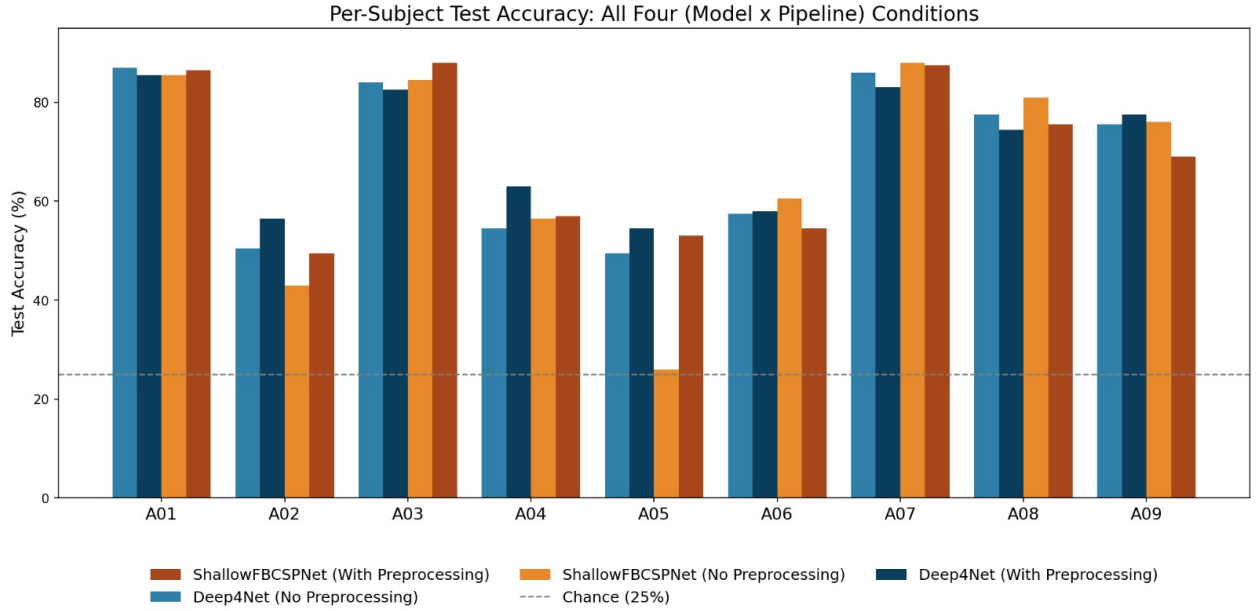


Fig. 5: All four (model, input) conditions overlaid on one per-subject accuracy axis. The four bars stay close on the efficient subjects and fan out on the difficult ones, where the input representation matters most; every configuration remains above the 25% four-class chance line for all subjects, with the ShallowFBCSPNet no-preprocessing pipeline on A05 the closest to it.

four MI classes well above chance from no-preprocessing signals, confirming that CNNs pick up useful spectral–spatial structure without hand-designed filtering. Even so, 8–30 Hz preprocessing continues to help, chiefly by steadying performance across subjects and rescuing the weakest, lowest-scoring ones, an effect in line with recent transfer-learning studies where alignment and domain-adaptation steps help most for subjects with a large distribution shift [18], [20]–[22].

The absolute accuracies here (about 67–71% on average) fall below the 78–85% frequently cited for recent CNN and convolution–transformer models on this data, such as EEG

Conformer [10], ADFCNN [8] and MSVTNet [11]. This shortfall is expected and largely procedural: such figures usually stem from within-session cross-validation, whereas I fit on session T and evaluate on the different-day session E after cross-subject pretraining, a stricter, more realistic arrangement that penalizes non-stationarity. Within this harder setting, the relative influence of the input representation is what I seek, and it is measured cleanly because every remaining factor is pinned down.

A few limitations deserve mention. I study a single dataset and two architectures and do not repeat runs over seeds. As the paired tests in Section V show, the small mean differences are

not significant at $n = 9$ and should be read as trends; the controlled effect I lean on is the tightened between-subject spread and the recovery of the weakest subjects. The preprocessing chain is sensible but not exhaustive; other bands or artifact-removal choices could tip the balance. Notably, Euclidean Alignment is present only in the Deep4Net pipeline and absent from the ShallowFBCSPNet pipeline, so the cross-architecture reading is confounded by alignment, and my controlled claim concerns the no-preprocessing-versus-preprocessed contrast within each architecture rather than a ranking of the two networks. Broader multi-dataset cohorts, repeated seeds, and applying alignment uniformly are the natural next steps toward a sharper comparison [13], [23].

VII. CONCLUSION

No Preprocessing EEG is enough for competitive four-class motor-imagery decoding with both Deep4Net and ShallowFBCSPNet. Preprocessing adds only a small mean accuracy gain but consistently cuts subject-to-subject variance and rescues the weakest subjects. In short: deep CNNs learn from no-preprocessing EEG, and preprocessing's main job is robustness, not peak accuracy. Future work: more datasets, multiple seeds, adaptive preprocessing.

DECLARATIONS

Data Availability: This work relies on the openly accessible BCI Competition IV dataset 2a; no fresh human-subject recordings were made, so no further ethical clearance was needed.

Author Contributions: The author contributed to the study design, implementation and analysis, and to the writing and review of the manuscript.

Conflict of Interest: The author declares no conflict of interest.

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